Mohammad Pashaei

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Research Interests

Multi-Agent Reinforcement Learning, emergent coordination, complex adaptive systems, and RL for sequential decision-making in partially observable environments.

Education

University of Tehran (UT) 2024–2026
M.Sc. in Sports Engineering, School of Engineering Science *Tehran, Iran*

- Focus: Applied Multi-Agent Reinforcement Learning & Complex Systems | GPA: 19.30/20 (4/4), top 3 in cohort
- Supervisors: Dr. Ali Fahim, Dr. Akbar Bohloul
- **Thesis:** *Opponent Tactical Awareness in Multi-Agent Reinforcement Learning for Football* (code)
- Selected coursework: Machine Learning (20/20), Deep Learning (20/20), Digital Image Processing (19.5/20), Statistical Data Analysis & Linear Algebra (19/20)

Isfahan University of Technology (IUT) 2018–2023
B.Sc. in Mechanical Engineering (Mechatronics) *Isfahan, Iran*

- **Thesis:** *Audio-Based Fault Diagnosis of Ventilation Fans Using Machine Learning Algorithms*

Conference Publications

- **Pashaei, M.**, Tayebi, A., Amiri, H., & Fahim, A. (2025). Short-term gains vs. long-term success: Reward strategy design for reinforcement learning in football. *Proceedings of the 32nd National and 10th International Iranian Conference on Biomedical Engineering (ICBME)*, 1327–1331, IEEE. DOI: 10.1109/ICBME68496.2025.11392257.
- **Pashaei, M.**, Forootan, E., Mahmoudi, S., Mazaheri, R., & Masoudi-Nejad, A. (2025). AXAN-Net: An adaptive exponential approximator neural network for latent feature composition with application in audio-based fan fault detection. *Proceedings of the 11th International Conference on Signal Processing and Intelligent Systems (ICSPIS)*, 474–479, IEEE. DOI: 10.1109/ICSPIS68676.2025.11551758.
- **Pashaei, M.**, Aghdaei, A., & Khorramabadi Arani, M. (2024). Audio-based fault diagnosis of ventilation fans using machine learning algorithms. *Proceedings of the 14th International Conference on Acoustics and Vibration (ISAV)*.

Research Experience

Graduate Researcher, University of Tehran Sport Engineering Lab 2025–Present
Advisor: Dr. Ali Fahim

- Developed a MARL-based football simulation environment to study opponent tactics, decentralized coordination, and emergent behavior.
- Investigated how behavioral rules and coordination norms emerge from agent interactions and environmental structure; extension of our previous research, presented at ICBME 2025.

Undergraduate Researcher, AVAlab, Isfahan University of Technology 2021–2022
Advisor: Dr. Saeed Ziaee Rad

- Developed machine-learning methods for ventilation-fan fault diagnosis from acoustic and vibration data, contributing to two conference publications (ICSPIS,ISAV).

Selected Research & Course Projects

- **Strategic Alignment & Reward Optimization in Multi-Agent Systems** (2026): Simulated multi-agent RL to study how cooperative behavior and effective rules emerge from environmental structure rather than top-down reward design (a finding of my master's thesis).
- **Adaptive Automated Insulin Delivery** (2026): Built a robust adaptive PID controller (EKF + ADMM) and benchmarked it against RL controllers (PPO, PPO-PID), achieving top time-in-range and zero hypoglycemia in virtual T1D/T2D trials. (link)
- **Multi-Objective Fuzzy C-Means Posture Classification** (2026): Used NSGA-II to optimize FCM thresholds, finding Pareto-optimal trade-offs between information preservation and cluster compactness. (link)

Technical Skills

- **RL & Optimization:** Stable Baselines3, Gymnasium, PettingZoo, pymoo, scikit-fuzzy
- **ML & Deep Learning:** PyTorch,PyTorch Geometric (PyG), JAX, scikit-learn, XGBoost
- **Data & Visualization:** NumPy,NetworkX, Pandas, SciPy, Matplotlib, Plotly
- **Programming:** Python, C++, MATLAB
- **DevOps & Tools:** Git/GitHub, Docker, Linux (Ubuntu), MySQL
- **Robotics / Embodied AI / Automation:** Arduino, Raspberry Pi, AVR, PLC programming, Factory I/O

Teaching Experience

Teaching Assistant, Digital Image Processing Mar–Aug 2026
University of Tehran (Instructor: Dr. Hadi Amiri)

- Designed homework assignments, course projects, and grading rubrics for a graduate class.(Material)

Teaching Assistant, Linear Algebra Sep 2025–Feb 2026
University of Tehran (Instructor: Dr. Hadi Amiri)

- Designed final projects and evaluation criteria.

Work Experience

Reliability Planner, SCI Feb–Aug 2024

- Formulated a sequential decision-making problem for maintenance scheduling under uncertainty; optimized workflows using data-driven reliability metrics.

Machine Learning Engineering Intern, Snowatec Jun–Aug 2022

- Built and deployed ML models for automated fault diagnosis using acoustic, vibration, and electrical-current signals; translated research methods into production pipelines.

Automation Engineering Intern, FarTest Jan 2022–Mar 2022

- Engineered and assembled an automated manufacturing line integrating PLCs, sensors, and actuators; improved production efficiency and system reliability.

Languages

English (fluent), German (beginner), Persian (native)